# Peter Samaan

Atlanta, GA | (678) 938-7787 | psamaan3@gatech.edu | U.S. Citizen | linkedin.com/in/peter-samaan

## Education

**Georgia Institute of Technology | Atlanta, GA** *January 2025 – Present*

Bachelor of Science in Electrical Engineering, Cumulative GPA: 3.39 Expected Graduation, May 2027

- Relevant Coursework: Signals and Systems, Intro to Signal Processing, Digital Design Lab, Advanced Programming Techniques for Engineering Applications, Engineering Software Design, Electromagnetics, Microelectronic Circuits

**Kennesaw State University | Marietta, GA** *August 2023 – December 2024*

Transfer with 69 Credit Hours, GPA: 3.72

- Relevant Coursework: Circuit Analysis, Digital Logic Design, Intro to Programming (C++), Probability and Data Analysis

## Skills

**Embedded & Hardware:** Microcontroller Development (Arduino, ARM mbed), FPGA Development (VHDL), Digital Logic Design, Oscilloscope Measurement, Logic Analyzer Measurement, Hardware-Software Integration, Soldering
**Signal Processing:** Digital Signal Processing (DSP), Digital Filter Design, Analog Filter Design, FFT Spectral Analysis
**Programming:** C/C++, Python (NumPy, SciPy), MATLAB, Verilog, VHDL, Assembly, R, Multithreading, OpenMP
**Engineering Software:** Quartus Prime, Circuit Simulation (LTspice, Multisim), AutoCAD Electrical, MS Office

## Experience

**Diversified Automation Inc. | Louisville, KY** *May – July 2025*

***Hardware Engineering Co-Op***

- Revised 200+ electrical schematics using AutoCAD Electrical to standardize documentation across multiple client contracts and ensure design compliance.
- Edited and updated technical hardware drawings across 5+ client contracts, improving accuracy of project documentation.
- Collaborated with engineers to update circuit layouts and electrical drawings, improving documentation accuracy for manufacturing and integration.

## Projects

**ECG Signal Processing Pipeline** *March 2026*

- Analyzed and processed ECG data from the MIT-BIH Arrythmia Database using DSP algorithms in Python (NumPy, SciPy).
- Designed digital IIR filters (Butterworth bandpass and 60 Hz notch) to remove baseline drift and powerline interference.
- Validated filtering performance in frequency domain using FFT spectral analysis.
- Developed R-peak detection algorithm to identify heartbeats and compute heart rate.

**Conway's Game of Life** *October 2025*

- Implemented "Conway's Game of Life" in C++ using sequential and parallel processing methods.
- Improved processing speed by 120% using both standard library threads and OpenMP threads.
- Added command line arguments to select processing type, window dimensions, and number of threads.

**Microcontroller Breakout Game** *April 2025*

- Developed an embedded system Breakout game in C++ on ARM mbed microcontroller, implementing UART communication to transmit game state data to an external LCD display.
- Implemented real-time input handling and game logic, debugging peripheral communication and timing behavior during hardware testing.

**FPGA LED Brightness Controller** *March – April 2025*

- Designed an FPGA-based LED brightness controller in VHDL, implementing a 128-level PWM to precisely control LED intensity.
- Implemented a gamma-correction lookup table (LUT) in VHDL to improve perceptual brightness accuracy.
- Developed an assembly-based embedded firmware demonstration that directly interfaced with the FPGA peripheral to dynamically adjust brightness across multiple LEDs.
- Verified system behavior using Quartus Prime simulation and hardware testing on an Intel DE-10 Standard board.

## Leadership

**Orthodox Chrisitan Campus Ministries (OCCM) | Atlanta, GA** *August 2025 – Present*

***Vice President***

- Organize and plan weekly edifying events for 50+ undergraduate and graduate students from 3 universities.
- Oversee monthly community service and outreach initiatives both on and off campus.

**St. Mark's American Coptic Orthodox Church | Doraville, GA**

***Youth Group Leader*** *August 2023 – Present*

- Engage class of 15+ students with activities, lessons, and service events to emphasize Orthodox Christian teachings and values.
- Foster connections between volunteers and students via outings and visitations.

***Worship Team Leader*** *November 2022 – Present*

- Engage congregation of 100+ with both contemporary and traditional worship music.
- Plan and lead monthly worship nights to encourage communal worship and prayer.